

Summer term 2026

Visual Data Analysis

Assignment Sheet 3

Solution has to be submitted via eCampus by May 1, 2026, 10:00 a.m.

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using [Jupyter notebook](#), please also export your scripts and results as PDF.

Exercise 1 (Parallel Coordinates Plot in Plotly, 11 Points)

Plotly is a plotting library with bindings for different programming languages. It enables the creation of interactive plots based on HTML, CSS and javascript. An introduction to the parallel coordinates plot with plotly can be found here: <https://plotly.com/python/parallel-coordinates-plot/>

In this exercises, we will work with a [Mice Protein Expression Dataset](#), which contains expression levels of 77 proteins, measured in the cerebral cortex of 8 classes of mice. The classes result from two genotypes (Ts65Dn, which serves as a mouse model of human down syndrome, vs. normal controls), two treatments (injection of the drug memantine vs. a saline solution as a control), and two experimental conditions related to context fear conditioning (context-shock, which should lead to learning, vs. shock-context, in which no learning takes place). Counting all repeated measurements, there are 1080 samples overall, some with missing data. You can find more information on the data in the [corresponding publication](#).

- a) Use pandas to read the file `Data_Cortex_Nuclear.xls` provided via eCampus. Extract subgroups t-CS-s and c-CS-s. How many mice were measured for each class? (1P)
- b) Create a parallel coordinates plot with plotly from the following 5 proteins: (pPKCG_N, pP70S6_N, pS6_N, pGSK3B_N, ARC_N). Assign different colors to the two selected classes. Annotate every axis with the correct protein name. (9P)
- c) Explore the data by interacting with the parallel coordinates plot. Do you find anything suspicious about the data set? (1P)

Exercise 2 (Comparing RadViz and Star Coordinates, 12 Points)

In the lecture, we introduced star coordinates and RadViz as alternative radial data visualization techniques. The paper `rubio-sanchez-radviz-sc-2016.pdf`, which is available via eCampus, provides a closer investigation of their relationship. Please read it and answer the following questions in your own words. Remember that **we will not grant even partial credit for copy-pasted text**.

- a) As we mentioned in the lecture, the star coordinates, and RadViz are two popular multidimensional data visualization techniques. Why they can also be interpreted as dimensionality reduction techniques? (1P)

- b) In general, the authors of this paper prefer star coordinates over RadViz. Briefly explain two reasons for their preference. (2P)
- c) Despite this, the authors mention a specific use case where they consider RadViz to be superior to star coordinates. Briefly explain what this use case is and why they prefer RadViz in this case. (2P)
- d) Neither RadViz nor star coordinates provide a one-to-one mapping between a data point's high-dimensional location and its two-dimensional projection. Interactively modifying the anchor points or axes, respectively, can reduce the resulting ambiguities. However, the paper mentions two examples in which, unlike star coordinates, RadViz introduces ambiguities that cannot be resolved with any re-arrangement of the anchor points. Point out these two examples. Could the extended RadViz that we learned about in the lecture resolve them? (3P)
- e) Several methods have been proposed to optimize the locations of anchor points in RadViz to obtain an improved separation of classes in the resulting projection. How did two such algorithms compare to supervised linear dimensionality reduction techniques in the experiments reported in this paper, in terms of (i) computational effort and (ii) achieved class separation? (2P)
- f) Briefly explain why anchor points in RadViz are commonly chosen such that they form a corner of the overall convex hull. Is this property also met for the star coordinate axis configuration in Fig. 12 (a)? Is it true in Fig. 12 (b)? (2P)

Exercise 3 (Principal Component Analysis, 17 Points)

In this task, you will use Principal Component Analysis (PCA) to explore a [Breast Cancer Dataset](#), which contains 699 samples, 458 of them indicating a benign tumor and 241 a malignant one. In the experiments, each measurement is graded between 1 to 10 at the time of sample collection. Nine measured characteristics such as clump thickness, uniformity of cell size, uniformity of cell shape etc. were found to differ the most between the benign and malignant samples. More details on the attributes and experiments can be found in the `breast-cancer-wisconsin.names` file.

- a) Read the `breast-cancer-wisconsin.xlsx` file. Note that there are some instances with missing data, which have to be imputed before we can run PCA. Pandas offers convenient functions for this. Apply an imputation method that makes sense for this dataset, and briefly explain your decision. (3P)
- b) Create a plot that, for any number n , shows what fraction of the overall variance in the data is contained in the first n principal components. Make sure that you only include the nine relevant numerical attributes in the PCA, not the sample codes or class IDs. How many components do we need to cover $\geq 90\%$ of the variance? (5P) *Hint:* You may use the implementation of PCA that is provided in the Python package scikit-learn.
- c) Each sample is now characterized by a point in PCA space. Create a scatter plot matrix that shows the first five principal components. Each diagonal cell should contain two overlaid density plots, one for the benign and one for the malignant class. Use different colors to distinguish between the classes, and add a legend that clearly states which samples are benign or malignant. (3P)
- d) Which PCA mode shows the strongest difference between the benign and the malignant samples? Name the original variables that have the highest and lowest weights in its definition, respectively. (3P)
- e) In the breast cancer dataset, all variables x_i have a similar range, $x_i \in [1, 10]$. If the variables of a dataset have very different ranges, for example one variable $x_1 \in [1000, 2000]$ and another one $x_2 \in [1, 5]$, how would this affect the PCA? Could it make sense to pre-process the data in such cases? Why and how? (3P)

Exercise 4 (Data Standardization Before PCA, 10 Points)

In this exercise, you will explore the effect of standardization on PCA using a subset of a simple two-dimensional [dataset](#) of height (inches) and weight (pounds) measurements of 100 different humans (`SOCR-HeightWeight_small.csv`). Then, you will discuss whether standardizing data beforehand makes sense in other scenarios.

- a) Create a scatter plot of the data. To ensure that distances in the plot reflect the distances that PCA accounts for, set the axis aspect ratio to one. Perform PCA on the data and plot the two principal components next to the original scatter plot, again with an axis aspect ratio of one. Also visualize the cumulative explained variance and the contributions of each variable to the principal components. (3P)
- b) Convert the given US units (inches and pounds) to metric units (m and kg). Re-create the same plots as in a) and describe your observations. (2P)
- c) Re-create the same two plots after standardizing the data to zero mean and unit variance in each attribute. How does this affect the PCA? Does it still matter which units have been used originally? (2P)
- d) Would you recommend standardization before PCA in this case? Justify your answer. (1P)
- e) For the following two scenarios, discuss the effects of standardization. Can you give a clear recommendation? Justify your answers:
 - Students rate the difficulty of a lecture and corresponding exercises on a scale of 1-10. For the lecture, most students give scores in the 4-6 range. For the exercises, some find them much more difficult than others, so we have a much wider spread of values between 1-10. We aim to compute a principal component that allows us to rank students with respect to how difficult they found the module overall. (1P)
 - The length of various bones is measured on a common scale (in cm) in healthy subjects and patients in which the growth of certain bones has been affected. We aim to compute an embedding in which patients form distinct clusters. (1P)

Good Luck!